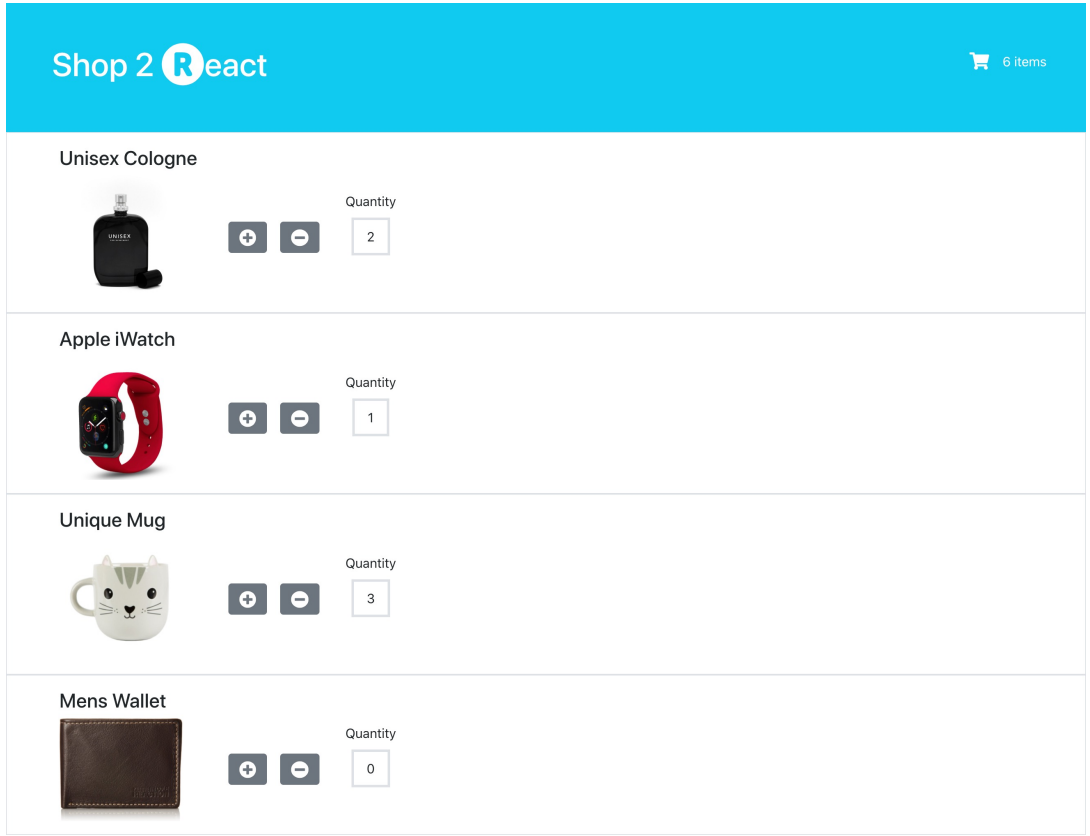
Lab Assignment

Lab Assignment

- In this week's assignment, you will continue building the basic shopping cart app from last week using React.
- Please make sure you use the files completed from the last lab.
- You are expected to utilize the concepts you have learned this week.
- The Todo List app from this week's homework has some similar principles that you can apply to this app.
- As a web developer, you are also expected to problem solve by doing your own research on concepts and techniques. Give yourself a 10 to 15 minutes to problem solve before asking for assistance.
- Like last week, you are also expected to do some additional research to meet some of the requirements of this assignment.

Lab Assignment

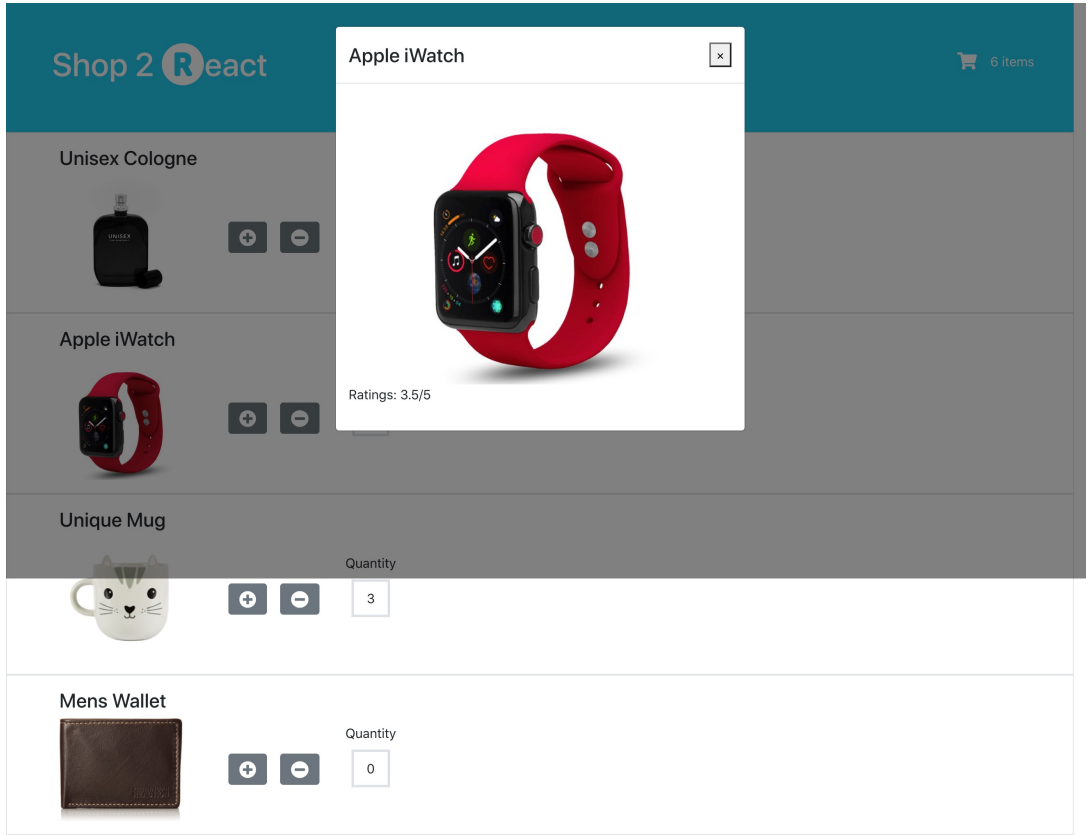
- Here's what the home of the app will look like at end of this lab assignment:



Note:
PDF also available for download on GAP.

Lab Assignment

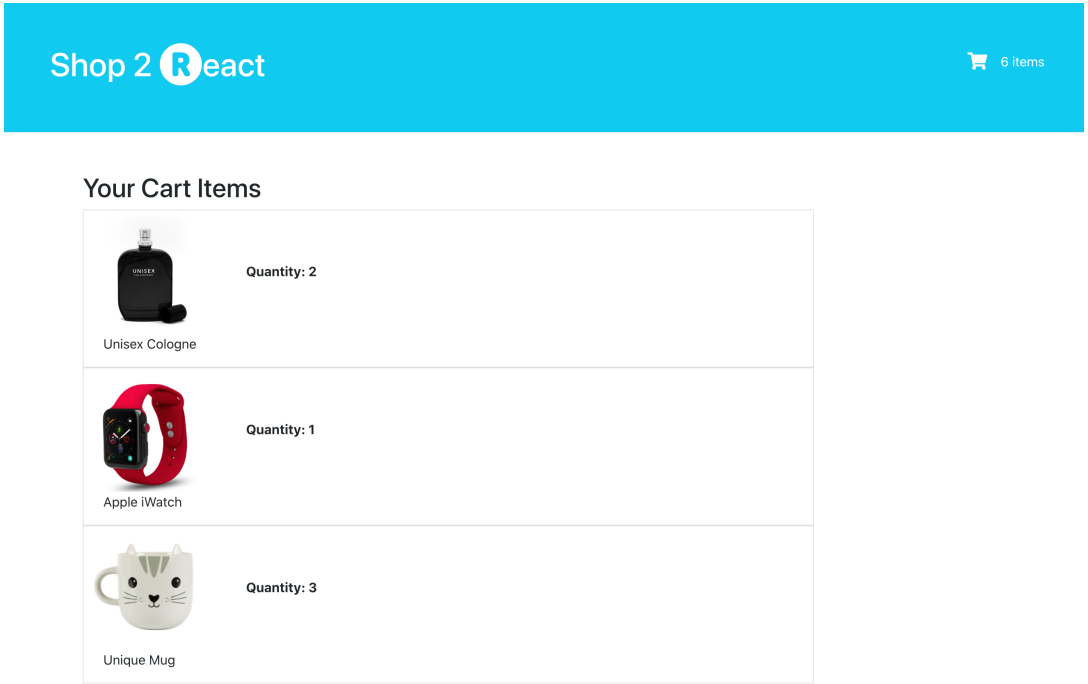
- Here's what the product popup will look like at end of this lab assignment:



Note:
PDF also available for download on GAP.

Lab Assignment

- And here's what the cart of the app will look like at end of this lab assignment:



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- Here’s some explanation how the app works:
  - Clicking on the product image, a lightbox will open displaying the product information. Clicking on the X button on the top right dismiss the lightbox.
  - Each add button click, quantity increases by 1.
  - Each subtract button click, quantity decreases by 1.
  - Total quantity of all products is display next to the cart icon which gets updated with each click.
  - Clicking on the shopping cart icon loads the cart screen which displays items that have 1 or more quantities.
  - Clicking on the shop logo returns to the home screen.

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- Here are your tasks breakdown:
  1. Separate components into its own js file. There should be 3 in the existing app that will be converted:
     - State data as products.js
     - Navbar as navbar.js
     - Display Products as displayProducts.js

     If you organized your components differently, consult your TA or professor on how to approach it.
  2. Add a modal (lightbox) to display the active product information: product description, image and a new property **ratings** set to a string number. Add this ratings property in the data first.
  3. Create add and subtract buttons in the display products that will call its respective handlers. These handlers should be on the App.js, below the constructor. You will need to use setState() in each handler. Refer to the Todos app for some guidance.

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- Continuation of tasks breakdown:
  4. Create a router in the navbar with links on the logo and cart icon. Make sure you install react-router-dom first. There will be 2 routes: home and cart, which both are js components. Because the home (Display Products) will now be called through the route, the original calling of Display Products will need to be removed in App.js. All data will be passed to the calling of the navbar component in App.js.
  5. Create the cart component as js file. In this component, you will bring in props pass from the route and dispersed using JSX to display the product information.
  6. Create icons using font awesome and style using react-bootstrap. You might need to add a couple/few custom css. You can do this in App.css or on a new css file.

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- Launch VS Code and browse to your shop app folder from last lab.
- To do task 2, make sure you have completed task 1 (separation of components).
- Open the **displayProducts.js** file. This file should have some imports at the top and then a standard function component with a props parameter. Before the return(), you will create a React Hook call useState(). This is new in React V16.8. (https://reactjs.org/docs/hooks-state.html) This state hook allow us to create whatever states we want in a standard functional component (which until now, you could only do that in a class component) including state data and other random data, like in our case: to show and hide modal (lightbox). This useState() hook want us to declare two things: a variable and a method ie:

```
const [variableName, methodName] = useState();
```

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- First, import **useState from React** library and **Modal from React-Bootstrap** (make sure you have this installed prior):

```
import { useState } from "react";
import { Modal } from 'react-bootstrap';
```

- Next, you will create the useState hook right at the top (before the *return()*) of your functional component:

```
const [show, setShow] = useState(false);
```

- The above state hook sets the show/hide of the modal.
- Below this hook, you will create 2 handlers using the state hook method to handle the calling from event clicks:

```
const handleClose = () => setShow(false);
const handleShow = () => setShow(true);
```

- Creating the handlers as arrow functions devoid us from having to bind them.

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- Within the return() – look for the product image within the map method. In the image, add this event attribute to call the show handler:

```
onClick={() => handleShow()}
```

- After the map method but still within the return(), you will create the modal component. Modal component comes with (just like table): header, body and footer. In the previous React-Bootstrap version, you will need to import each separately but on the new version, you only need to import the main Modal. Here's the initial Modal you will set up after the map method:

```
<Modal show={show} onHide={handleClose}>
    <Modal.Header closeButton>
        <Modal.Title></Modal.Title>
    </Modal.Header>
    <Modal.Body>
    </Modal.Body>
</Modal>
```

Note: *closeButton* is a boolean attribute that comes bundled with Modal. You don't need to create this X button separately!

Here's more Modal examples you can check out: https://react-bootstrap.github.io/components/modal/

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- This Modal when opens will display the product information such as description (title), same image but larger and the ratings (out of 5). Before proceeding, head to your products data (products.js) and add a 4th property call ratings for each product. You can insert anywhere but I suggest between desc and value. Example: ratings: “3.5”.
- Now that is set up, you can now add a 2nd state hook for the product:

```
const [show, setShow] = useState(false);
const [showImge, setShowImge] = useState({});
```

- The product within the map method needs to be passed to the show handler, so modify the calling of the handler to:

```
onClick={() => handleShow(product)}
```

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- Let's head back to the handlers at the top. Modify the show handler as follows:

```
const handleClose = () => setShow(false);
const handleShow = (product) => {
    setShow(true);
    setShowImge(product);
}
```

- With the handlers now ready, you will now head back to the Modal component to insert the product information. Let's start with the header:

```
<Modal.Header closeButton>
    <Modal.Title>{showImge.desc}</Modal.Title>
</Modal.Header>
```

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- Next – the body. You will also add the ratings in here:

```
<Modal.Body>
    <img
        src={showImge.image}
        width="350"
        alt={showImge.desc}
        className="mx-5"
    />
    <p>Ratings: {showImge.ratings}/5</p>
</Modal.Body>
```

- Last but not least, add some styling to the ratings:

```
<p><span className="text-dark">Ratings:</span> {showImge.ratings}/5</p>
```

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- Test out the Modal on the browser (npm start). When a product image is clicked on the home screen, it should open the Modal displaying the 3 pieces of information of the product (refer to the example in the beginning of this presentation). If it didn't work or display correctly, try go back to the previous slides(s) and try resolve on your own first. If after 10 minutes unsuccessful attempt, consult your TA.
- If everything works accordingly, you may proceed to the rest of the tasks. Click here to return to that slide.
- Happy coding and, good luck!

Lab Assignment

- **DUE:**

  Today, 10.30PM PT.

- **Submission:**

  Post Github URL (with commit #2) on GAP Week 2 Day 3 Lab Assignment

- **Quiz Reminder:**

  Take your quiz while in the lab today. Close tomorrow (Sunday) at 11.59PM.

Connect with Us (WEB403/603/803)

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End of Presentation